

Andrea Zingarofalo

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📍 Brussels, Belgium

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Education

PhD	Université libre de Bruxelles , Quantum Information Theory	Brussels, Belgium
	- Thesis: Self-Testing and Device-Independent Verification of Quantum Computation - Advisor: Dr. S. Pironio	Sept 2025 – present
Diploma	Advanced School ISUFI (Excellence Program) , Natural Sciences	Lecce, Italy
Li-cenza	70/70 cum laude — ~10% acceptance rate	Sept 2020 – July 2025
Exchange Program	École normale supérieure , Theoretical Physics - Master ICFP M2	Paris, France
		Sept 2024 – July 2025
MSc	University of Salento , Theoretical Physics	Lecce, Italy
	110/110 cum laude - Thesis: Exploring the boundaries of quantum correlations: Bell inequalities, convex duality, and self-testing - Advisors: Prof. L. Martina & Prof. J.-D. Bancal	Sept 2023 – July 2025
BSc	University of Salento , Physics	Lecce, Italy
	110/110 cum laude - Thesis: Study of one-dimensional quantum integrable systems via the Bethe Ansatz - Advisor: Prof. L. Martina	Sept 2020 – July 2023

Experience

Laboratoire d'Information Quantique - Université libre de Bruxelles , Doctoral Researcher	Brussels, Belgium
- Research on self-testing and device-independent protocols for certifying components of quantum computation, from multipartite entanglement to error-correcting codes - Supervisor: Dr. S. Pironio	Sept 2025 – present 8 months
Institut de Physique Théorique - CEA Paris-Saclay , Visiting PhD Student	Paris, France
- Research on verification protocols for quantum computation - Supervisor: Prof. J.-D. Bancal	Oct 2025 – Feb 2026 5 months
Mindrift , AI Model Training Specialist (Physics)	Remote
- Contributed to training and evaluation of AI models by providing expert-level physics supervision and structured feedback - Analyzed model outputs to identify systematic failure modes and guide iterative model refinement	Nov 2025 – present 6 months
Institut de Physique Théorique - CEA Paris-Saclay , Research Intern	Paris, France
- Derived analytically new Bell inequalities for partially entangled states - Implemented SDP relaxations to numerically estimate quantum bounds of Bell inequalities via the Navascués-Pironio-Acín hierarchy - Supervisor: Prof. J.-D. Bancal	Mar 2025 – July 2025 5 months

Mathematical Physics Group - University of Salento (with University of Glasgow),
Research Intern

- Derived multi-kink solutions of a generalized Fokker-Planck equation via Hirota's bilinear method
- Developed automated Mathematica scripts for symbolic generation of soliton solutions
- Supervisor: Prof. L. Martina

Lecce, Italy
Mar 2024 – July 2024
5 months